

Maleen Sathsara

Cyber Security Undergraduate

Phone : +94 77 0278 262

Email : maleens.rox@gmail.com

Address : Colombo, Sri Lanka

GitHub : [maleenrox](#)

LinkedIn : [maleen-sathsara](#)

Portfolio : [maleen.dev](#)

Professional Summary

Cybersecurity undergraduate specializing in network defense, threat intelligence, and security operations. Highly analytical and skilled in Python, C, Rust, and Bash scripting, with hands-on experience developing command-line security tools and advanced networking projects. Proven ability to research emerging vulnerabilities and evolving attack vectors to document technical findings and security reports. Equipped with strong problem-solving capabilities, excellent communication skills, and a collaborative mindset tailored for team environments. Actively seeking the Intern - Cyber Security role to assist in configuring firewalls, managing IDS/IPS solutions, executing health checks, creating standard operating procedures (SOPs), and contributing to cross-functional IT initiatives with proactive defense strategies.

Education

BSc (Hons) in Information Technology Specializing in Cyber Security | 2024-2028 (Undergraduate)

Sri Lanka Institute of Information Technology (SLIIT), Malabe, Colombo.

- Program Structure : General IT Modules (1.5Y), Advanced Cyber Security Specialization (2.5Y).
- General Coursework : Covering Software Programming, Networking, AIML, Databases, Advanced Math.
- Specialization Coursework : Network Security, Vulnerability Assessment & Penetration Testing, Ethical Hacking & Cyber Ops, Digital Forensics, Cryptography, Governance Risk & Compliance.
- Current Cumulative GPA : 3.53

Higher Diploma in Computer Science | 2018-2020 (Completed)

Kosmotec International Information Technology Training Centre, Homagama, Colombo.

- Relevant Coursework : Digital Documentation (MS Office), Graphic Designing, Programming, Hardware.
- Final Score : Distinction Plus (A+).
- Noticeable Awards : Best IT Award Winner 2019 and 2020.

Technical Skills

- Security & Network: Firewalls Basics, IDS/IPS Systems, Antivirus Solutions.
- Cybersecurity Concepts: Security Tool Development, Vulnerability Assessment, Threat & Attack Vector Research, Security Health Checks, Patch Management, Firmware Upgrades, Log Analysis.
- Networking: Advanced TCP/IP, VLANs, Inter-VLAN Routing, VLSM, Traffic & Protocol Analysis.
- Programming & Scripting : Python, Rust, C, Java, JavaScript, TypeScript, R, Bash/CMD, VB, Assembly.
- Security & Dev Tools : Git, CLI Tools, Linux (Kali, Ubuntu), JetBrains IDEs, Postman, Wireshark.
- Documentation & Collaboration: Standard Operating Procedures (SOPs), Configuration Guides, Security Reports, Cross-functional Team Collaboration.

Certificates and Licenses

Introduction to Cyber Security - CISCO | 2025

CISCO Network Academy

- Relevant Coursework : Introduction to Cyber Security, Attacks / Concepts and Techniques, Protecting Data / Privacy / Organizations, Future of Cyber Security.

Ethical Hacker - CISCO | 2026 (In Progress)

CISCO Network Academy

- Relevant Coursework : Introduction to Ethical Hacking and Penetration Testing, Planning and Scoping a Penetration Testing Assessment, Information Gathering and Vulnerability Scanning (Pending).

Projects & Research

Ripper Fence 1.0 - Secured Password Generating Tool (Personal Research)

Researched and Developed a Python-based cross-platform password generator using cryptographically secure randomness to mitigate weak password risks and improve credential security.

- Language : Python 3
- Core Modules : secrets, string, os, sys, subprocess, platform, pathlib, datetime, gc
- Key Features : Cryptographic randomization, CLI UX, Cross-platform (Windows/Linux/macOS), Secure password file saving options, Clipboard integration.
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Shadow Markets and Stealth Vectors - Cybersecurity Research Paper

Researched and authored a comprehensive cybersecurity analysis report detailing the atomic-level mechanics of Living off the Land (LotL) attacks and their dark web commodification.

- Core Focus : Fileless in-memory execution, native OS binary exploitation (PowerShell/Bash), and stealth evasion mechanisms.
- Concepts Analyzed : Process Hollowing, Reflective DLL Injection, Ransomware-as-a-Service (RaaS), and Tor/PGP infrastructure.
- Methodology : Threat intelligence synthesis, technical case study analysis (MGM/Caesars breaches), and EDR countermeasure evaluation.
- Research Standard : Compliant with IEEE academic formatting and referencing guidelines.

Secure Concurrent TCP Application Server

Engineered a secure, concurrent TCP client-server system featuring custom protocol framing, salted hash authentication, session tokens, and brute-force protection.

- Languages : C (Server implementation) , Python 3 (Client application)
- Core Modules : sys/socket.h, sys/wait.h, arpa/inet.h, fork(), socket
- Key Features : Explicit TCP framing architecture , Concurrent process management, Time bound cryptographic session tokens , Persistent audit logging of client commands and states.
- Concepts : Network Security, Concurrency, Socket Programming, Application-layer protocol design.

Enterprise Network Infrastructure Implementation & Analysis

Designed and configured a multi-site WAN network with VLAN segmentation, 802.1Q trunking, inter-VLAN routing, and RIPv2 dynamic routing across VLSM subnets. Validated TCP/IP and application-layer protocols through network traffic and PDU analysis.

- Simulation Tool : Cisco Packet Tracer, Wireshark
- Switching & Routing : VLANs (802.1Q), Inter-VLAN Routing, Trunking, RIPv2, Static Routing
- Protocols Analyzed : TCP/UDP, HTTP, FTP, DNS, SMTP/POP3, ARP, ICMP, IPv4/IPv6
- Network Design : VLSM (Variable Length Subnet Masking), WAN Serial Connections

AutoLux Car Rental Management System - Full Stack (Side project)

Developed a full-stack car rental management system with role-based authentication, CRUD functionality, analytics dashboards, and a responsive UI.

- Frontend : Angular, TypeScript, HTML, CSS, RxJS, ChartJS
- Backend : Java, Spring Boot, Lombok, JPA Hibernate
- Database : MSSQL (Microsoft SQL Server) - SQL Express
- Tools : IntelliJ IDEA, WebStorm, MS Server Management Studio, Postman, GitHub

Soft Skills

- Deep Analytical Thinking & Problem Solving.
- Adaptability & Continuous Learning.
- Cross-Functional Teamwork & Leadership.
- Effective Technical Communication.
- Attention to Detail & Ingenuity.

Additional Information

- Languages: English (Fluent), Sinhala (Native).
 - Availability: Immediate joiner.
 - Work Authorization: Eligible to work in Sri Lanka.
 - Shift Flexibility: Willing to work rotational and night shifts.
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